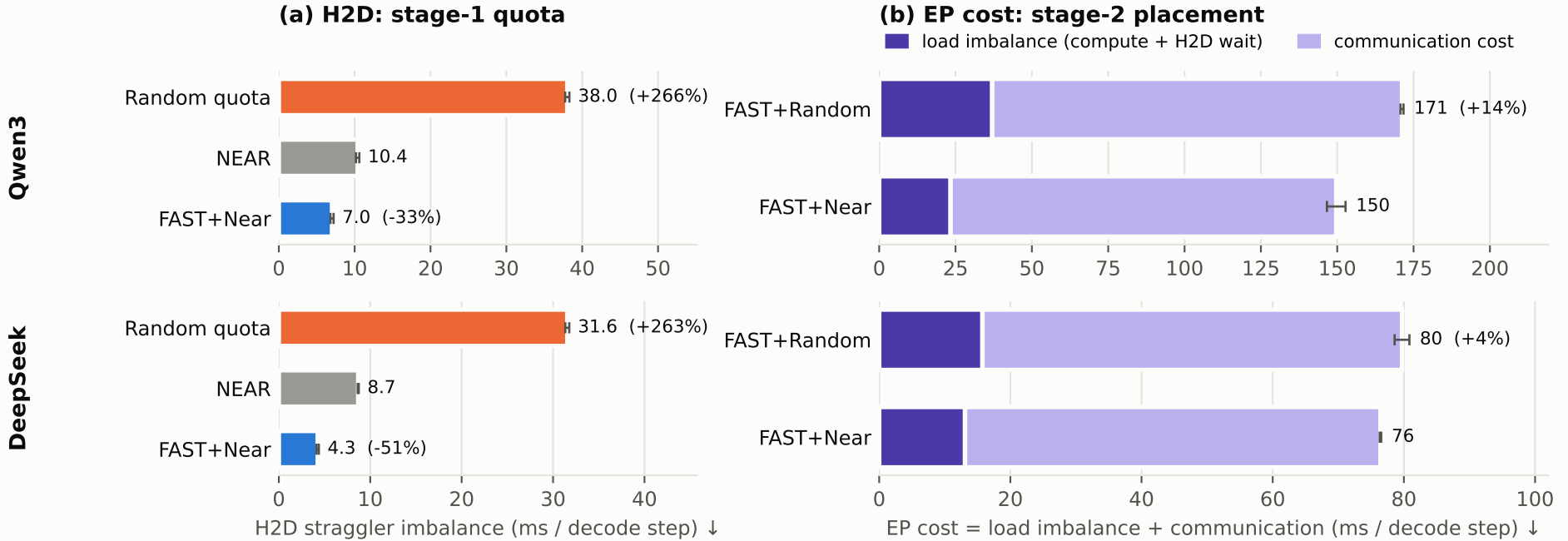
Stage-wise cost breakdown: R8, serial expert execution, inline H2D (per decode step, mean ± std, 3 runs)



(a) relative to NEAR. Random quota: unbalanced (i.i.d. random rank per miss) + Near placement. (b) relative to FAST+Near (same FAST quota, stage-2 placement only).